Hybrid-RIS Empowered UAV-Assisted ISAC Systems: Transfer Learning-Based DRL

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***Abstract*— In this paper, we consider a novel hybrid reconfigurable intelligent surface (HRIS) consisting of active as well as passive reflecting elements mounted on unmanned aerial vehicle (UAV). The aim is to improve air-to-ground communication by assisting multiple users, while detecting several low mobility targets. We formulate a sum-rate optimization problem that accounts for statistical channel estimation errors (SCEEs) to concurrently fine-tune both active and passive phase-shift matrices, UAV trajectory, and transmit beamformer for integrated sensing and communication (ISAC). Subsequently, we introduce a transfer learning based approach combining with deep deterministic policy gradient (DDPG) to enhance the overall data rate while minimizing the time it takes for users to transmit data. Additionally, we present an alternating optimization (AO) algorithm that employs a repetitive method to address the combinatorial nonconvex optimization problem and offers a solution that is very close to optimal. Finally, we showcase the superiority of the proposed scheme through Monte Carlo simulations. Also, we have compared the performance with perfect channel state information (CSI) counterpart. The outcomes of simulations confirm the theoretical analysis and demonstrate the efficiency of the proposed framework. Additionally, the results reveal the advantages of incorporating HRIS aided UAV assisted ISAC in improving the quality of both communication and sensing performance.**

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***Index Terms*— Estimation error, hybrid reconfigurable intelligent surface (HRIS), integrated sensing and communication (ISAC), sum rate and unmanned aerial vehicle (UAV).**

I. Introduction

INTEGRATED sensing and communications (ISAC) is becoming a pivotal tool for the evolution of next generation networks [1]. ISAC's primary characteristic is its capability to utilize the same resources for both communication and sensing in terms of time, frequency and power [2], [3], [4]. The performance of ISAC systems in sensing and communication (S&C) is quantified using the following metrics: i) the sensing rate (SR) [5], [6], [7], and ii) the communication rate (CR). However, using radar signals for both communication and sensing often results in reduced performance, employing radar waveforms for ISAC can limit the communication data rate. Recent advancements in ISAC involve designing beamformers to ensure a minimum signal-to-interference-plus-noise ratio (SINR) for users while minimizing the radar sensing error [8]. A radar priority strategy has been suggested to maximize radar signal-to-noise ratio (SNR) while maintaining a minimum SINR for as many users as possible [9]. The use of wider bandwidths in millimeter wave (mmWave) provides the opportunity to attain higher data rate and enhance radar range precision for ISAC [10]. However, at higher frequencies, signals become more susceptible to obstructions, resulting in a significant degradation in both sensing and communication performance. To address these issues, the use of reconfigurable intelligent surfaces (RISs) is gaining attention [11], [12], [13], [14], [15], [16].

Recent studies [17], [18], [19], [20], [21], [22], [23] have explored the integration of RIS into ISAC systems. This integration aims to enhance per-user SINR and/or overall target illumination power, particularly in scenarios where the direct path from the distributed focal beam steering to users or targets are obstructed. Additionally, RIS has been exclusively utilized to enhance communication capabilities when the direct links between transceivers and targets are established for sensing [21], [22], [23]. It has also been used to enhance sensing by establishing a virtual link between the Base Station (BS) and the target. In [23], the authors worked on optimizing the beampattern directed at the target, ensuring enhanced power illumination on the target for improved sensing capabilities. This is achieved by jointly optimizing

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the active and passive beamforming matrices [24]. In [25], the authors focused on hybrid RIS (HRIS)-aided millimeter wave (mmWave) OTFS systems, which are particularly suited for high-mobility communication environments. The paper introduced a joint channel estimation and data detection (JCEDD) scheme designed for orthogonal time frequency space (OTFS) modulation. However, [26] presented a simultaneous transmission and reflection RIS (STAR-RIS)-aided ISAC scheme designed for vehicle localization and communication. This work specifically integrates sensing capabilities with communication by placing a STAR-RIS on the outside surface of the vehicle, facilitating both communication with in-vehicle UE and sensing through nearby roadside units (RSUs). Both papers emphasize the use of RIS in high-mobility scenarios to address challenges posed by Doppler shifts and time-varying channels. In contrast to these works, our research focuses on low-mobility users/targets within a RIS-aided ISAC system. While these high-mobility solutions target fast-moving vehicles and rapid channel variations, our framework is designed to optimize communication and sensing for users with low mobility. This distinction allows for a different set of design parameters, as the challenges associated with doppler shifts are less prominent. Instead, the focus of our work is to improve system performance for low mobility users, where stability and accuracy of communication and sensing take precedence.

Recently, there has been substantial exploration in the field of unmanned aerial vehicles (UAVs) [27], [28], [29], [30], [31]. The authors of [27] aimed to increase the throughput of a single UAV assisted single-cell multi-user network utilizing orthogonal frequency division multiple access (OFDMA). Further, a multiple UAV assisted cellular network was considered in [28], where UAV-to-UAV transmission pairs utilized the same spectrum with the uplink of terrestrial cellular users. However, UAV networks face several challenges that hinder their efficiency [31]. Issues, such as links being obstructed, and the amplification of signals by UAVs leading to higher power usage, which in turn impacts the UAV's longevity. Due to the advantages RIS offers, there has been widespread research on RIS assisted UAVs networks [32], [33]. For instance, the investigation conducted in [33], presented two approaches to optimize both spectral efficiency (SE) and EE in RIS assisted UAV network. While the aforementioned studies highlight RIS-assisted UAV benefits, they often overlook the channel estimation challenges for passive RISs. This is due to the "double fading" phenomenon observed in cascaded channels. To counterbalance this challenge of passive RIS and also improving the SE, recent studies explored fully active RIS [34] which limits the EE.

To address the shortcomings of both fully passive [13], and active RISs [43], [44], hybrid RIS (HRIS) was suggested in [39], [45], [46], [47], [48], and [49]. By incorporating signal amplification in active components, the HRIS can efficiently compensate for the cascaded path loss (PL) and also improves SE and EE. The authors of [40] focused on maximizing target illumination power and ensuring optimal SINR for communication, while controlling noise from the H-RIS's active elements. Moreover, the authors of [35] discussed on robust beamforming for a hybrid RIS-assisted ISAC system, considered target location estimation error to optimize the sensing gain while meeting SINR and power constraints, using an alternating optimization (AO). The study in [36] explored the balance between active and passive elements in a HRIS to maximize EE. The authors of [37] presented a hybrid RIS-enhanced UAV communication system that aims to boost user rates by jointly fine-tuning the UAV's position, power distribution, and the RIS's reflective and amplifying properties, overcoming the inherent nonconvex optimization challenges through block coordinate descent and successive convex approximation (SCA) techniques. Nevertheless, the aforementioned studies sought a closed-form solution for the joint trajectory and phase shift problem in RIS-assisted UAV systems, employing methods like SCA. However, these solutions are computationally complex and challenging to implement in real-time due to the dynamic nature of UAV system states. To address these issues, authors in [38], [41], and [42] employed deep reinforcement learning (DRL) algorithms like double-deep Q-learning (DDQN) and deep-deterministic policy gradient (DDPG) for real-time optimization of UAV trajectory and passive beamforming. The authors of [38] investigated the performance of a secure multiple input single output (MISO) satellite communication system enhanced by a HRIS, focusing on a joint beamforming design to maximize secrecy rates while accounting for outdated CSI and DRL based power utilization. However, these studies only optimized the movement of UAVs in azimuthal direction. Additionally, they focused on a uniform linear array (ULA) for the RIS phase-shift, rather than a more realistic uniform planar array (UPA) [50].

A. Motivation and Contribution

Unlike existing literature as in Table. I, in this study, both the UAV and targets are treated as mobile rather than stationary. Consequently, the HRIS-UAV network under consideration is inherently dynamic, posing challenges for traditional optimization methods. Further, the objective of deploying and designing the UAV-RIS system is to maximize long-term advantages rather than immediate gains. The objective of this work is to maximize the achievable sum rate through the joint optimization of phase shift matrices, ISAC beamformer, and UAV trajectory. By optimizing these components collectively, the goal is to enhance the overall communication performance of the system. This places the focus in the realm of reinforcement learning (RL) algorithms [51], as these algorithms are capable of considering the foreseen evolution of the system instead of narrowly focusing on optimizing immediate benefits. Most of the literature on RIS-assisted ISAC systems have considered the scenario of perfect channel state information (CSI). However, it is challenging to acquire the precise target location or the instantaneous CSI. Also, UAVs equipped with RIS can enhance both communication and sensing by providing better communication links for data transmission and improved sensing capabilities via better signal control. This necessitates the need for a robust design optimization in HRIS empowered

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TABLE I
CONTRASTING OUR WORK WITH STATE-OF-THE-ART LITERATURE

	[33]	[34]	[35]	[36]	[37]	[38]	[39]	[40]	[41]	[42]	**Our work**
Hybrid RIS	X	X	✓	✓	✓	✓	✓	✓	X	X	✓
UAV	✓	X	X	X	✓	X	X	X	✓	✓	✓
Perfect CSI (pCSI)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Imperfect CSI (iCSI)	X	✓	✓	X	X	X	X	X	X	X	✓
Sum-rate maximization	✓	X	X	X	X	X	✓	X	X	X	✓
ISAC	X	✓	✓	X	X	X	X	✓	X	X	✓
DRL	X	X	X	X	X	✓	X	X	✓	✓	✓
Capon beamformer	X	X	X	X	X	X	X	X	X	X	✓
Optimization approach	✓	✓	✓	✓	✓	X	✓	✓	X	X	✓

UAV-aided ISAC systems using a novel RL [52] approach, i.e., transfer learning enhanced DDPG, comprising of multiple non-stationary targets and multiple users under imperfect CSI consideration.

The motivation of implementing DDPG based transfer learning is to reduce the time and computational resources required for training. By leveraging a pre-trained network, we bypass the time-consuming phase of learning from scratch. The transferred network, which inherits weights from a well-trained model, is likely to perform better right from the start compared to a network with randomly initialized weights. This head start often leads to enhanced overall performance, as the network builds upon the learned patterns and features from the initial task. Motivated by the aforementioned research, we focus on a HRIS mounted on UAV, where a dual functional base station (DFBS) communicates with several users and simultaneously senses multiple low mobility vehicular targets. The novel key contributions are:

- We formulate an optimization problem while considering statistical channel estimation error (SCEE) to concurrently fine-tune both active and passive phase matrices, UAV trajectory, transmit beamformer for sensing and communication to maximize the achievable sum-rate.
- Given the non-convexity of the formulated problem, traditional analytical optimization methods prove highly challenging to apply, as they require unconventional and intricate approximations. Consequently, we utilize a DDPG based transfer learning approach offering the opportunity to enhance a RL agent by leveraging insights and expertise from a relevant task.
- The achievements of DRL and transfer learning (DTL) in managing resources have motivated us to integrate both. We present a framework based on DDPG and transfer learning to enhance the overall data transmission rate while minimizing the time it takes for users to transmit data. This approach can efficiently and rapidly train a new agent to address the optimization problem.
- Moreover, Capon beamforming has been utilized for the formulated DRL problem where interference sources may change over time. In designing a system where the base station has dual functionality, it's essential to ensure that the sensing task cause no interference to the communication system.
- Additionally, for HRIS assisted UAV ISAC systems, we use the Gaussian-Kronecker model for SCEE, validated by minimum mean square error (MMSE). For baseline comparison, we delve into the solution structures by utilizing AO technique. For transmission matrices, we use the Lagrange multipliers, and for HRIS phase-shift vectors and trajectory of the UAV, SCA is applied.
- Finally, we demonstrate the superiority of the proposed system over the benchmark schemes, such as passive RIS, random phase, no-RIS case and DDPG. Additionally, we illustrate how the number of reflecting elements and SCEE parameters influence the rate performance.
- Finally, the beampattern gain across various azimuth and elevation angles is illustrated.

TABLE II
KEY NOTATIONS

Notation	**Definition**	**Notation**	**Definition**
$\mathbf{A}^{\star}$	Conjugate of $\mathbf{A}$	$\mathbb{C}^{M\times M}$	Complex matrix
$\mathrm{diag}(\cdot)$	Diagonalization operator	$(\mathbf{A})^{\mathrm{H}}$	Hermitian
$(\mathbf{A})^{\mathrm{T}}$	Transpose	$\mathrm{Re}\{\cdot\}$	Real part
$\lvert\mathbf{A}\rvert$	Modulus operator	$\mathbb{C}^{M\times 1}$	Complex vector

Structure of the Paper: The remainder of this paper is structured into five main sections. Section I provides an introduction, Section II describes the system model and Section III illustrates the designing of DRL in H-RIS assisted UAV environment. Section IV refers to integration of DDPG with transfer learning and Section V outlines the optimization solution. The effectiveness of our approach is showcased in Section VI through simulation results. The paper wraps up with main remarks in Section VII. Moreover, the key notations are enumerated in Table II.

II. SYSTEM MODEL

We consider a HRIS assisted UAV system, in which the HRIS is mounted on a UAV to assist communication for U users (UE), while detecting K targets (TE) to enhance air-to-ground communications [53], as shown in Fig. 1. The UE and TE have $N_u = N_k = N$ antennas. For target detection, we partition the detection area into D rows and F columns. Each column represents a segment of width $\Delta x = \frac{x_{max}-x_{min}}{F}$. Similarly, each row represents a segment of width $\Delta y = \frac{y_{max}-y_{min}}{D}$. If the target k is in the jth row, then its y-coordinate would satisfy $y_{min} + (j-1)\Delta y \leq y_k < y_{min} + j\Delta y$. If the target k is in the ith column, then

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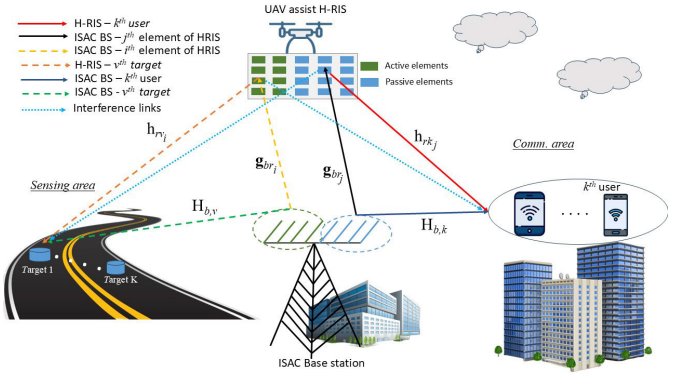


Fig. 1. System model.

its x-coordinate would satisfy $x_{min} + (i-1)\Delta x \leq x_k < x_{min} + i\Delta x$. Each segment can reflect signals from both the BS and the HRIS. The reflected signals are then captured by the HRIS. The BS is equipped with M antennas and the HRIS consists of L elements, which includes L_a active and L_p passive reflecting elements such that $L = L_a + L_p$ and $1 \leq L_a \leq L_p$. The active elements are implemented using low-power reflection amplifiers [34] and the fully passive RIS with $L_a = 0$ representing a specific instance of the HRIS. Let $\boldsymbol{\theta} \triangleq \text{diag}(e^{j\beta_1^{pas}}, \cdots, e^{j\beta_{L_p}^{pas}})$ represent the reflection matrix for passive elements and for the active elements, $\boldsymbol{\phi} \triangleq A_a \boldsymbol{\phi}_a$ such that the phase-shift matrix is defined as $\boldsymbol{\phi}_a \triangleq \text{diag}(e^{j\beta_1^{ac}}, \cdots, e^{j\beta_{L_a}^{ac}})$, while the amplification matrix as $\boldsymbol{A}_a = \text{diag}(\alpha_1, \cdots, \alpha_{L_a})$.

Let $\mathbf{g}_{br_i}[n]$ and $\mathbf{g}_{br_j}[n] \in \mathbb{C}^{1\times M}$ denote the channel gain between the BS and i^{th}, j^{th} elements of HRIS, respectively, $\mathbf{h}_{rt_i}[n] \in \mathbb{C}^{N\times 1}$ represents the channel gain between the ith element of HRIS-TE and $\mathbf{h}_{ru_j}[n] \in \mathbb{C}^{N\times 1}$ represents the channel gain between the jth element of HRIS-UE. The direct connection from the DFBS to the TE and the UE are denoted as $\mathbf{H}_{b,k} \in \mathbb{C}^{N\times M}$ and $\mathbf{H}_{b,u} \in \mathbb{C}^{N\times M}$, respectively. Without loss of generality, the three dimensional (3D) coordinates of the BS, UAV and UE are $\Xi_{BS} = [x_{BS}, y_{BS}, 0]^{\mathrm{T}}$, $\Xi_{UAV}(t) = [x_{UAV}, y_{UAV}, z_{UAV}]^{\mathrm{T}}$ and $\Xi_{UE} = [x_{UE}, y_{UE}, 0]^{\mathrm{T}}$, respectively. For simplicity, the time period is divided into $\mathcal{N}$ segments, such that $T = \mathcal{N}\Delta T$. The UAV flies at a speed v_n in a particular direction κ_n during time slot $n \in 1, \cdots, \mathcal{N}$. The following conditions are considered for n-th time slot:

$$\varpi_{UAV}[1] = \varpi_{UAV}[\mathcal{N}], \tag{1}$$

$$\|\varpi_{UAV}[n+1] - \varpi_{UAV}[n]\|^2 \leq D^2, \quad n = 1, \ldots, \mathcal{N}-1. \tag{2}$$

The above represents the UAV's movement and $D = v_n \Delta T$. The trajectory of the UAV is $\iota = \{\Xi_{UAV}[1], \cdots, \Xi_{UAV}[\mathcal{N}]\}$. Thus, the distances d_{bu} between the BS and UAV and that between the UAV and UE(d_{uu}) are given as

$$d_{bu}[n] = \sqrt{|\varpi_{UAV} - \varpi_{BS}|^2 + z_{UAV}^2}, \tag{3}$$

$$d_{uu}[n] = \sqrt{|\varpi_{UAV} - \varpi_{UE}|^2 + z_{UAV}^2}. \tag{4}$$

The communication between the UAV and the ground receiver is influenced by factors such as the angle of elevation (AOE) and the surrounding environment. The AOE at UAV, and that at the UE are expressed as

$$\varphi_{UAV} = \arctan\left(\frac{z_{UAV}}{\varpi_{UAV} - \varpi_{BS}}\right), \tag{5}$$

$$\varphi_{UE} = \arctan\left(\frac{z_{UAV}}{\varpi_{UAV} - \varpi_{UE}}\right). \tag{6}$$

Subsequently, the probability of having LoS is determined as $P_{\text{LoS}}(\varphi) = \frac{1}{1+b\exp(-a[\varphi-b])}$, where a and b are the constant environmental parameters, and φ is the AOE at the ground nodes. PL exponents for aerial communication links are $\zeta(\varphi) = P_{\text{LoS}}(\varphi)\zeta_e + \zeta_0$ such that ζ_e and ζ_0 are constant. In practical scenarios, it is commonly observed that the link between the UAV and ground UEs poses LoS with a high likelihood. Therefore, we assume free-space PL model for the communication channel between the UAV and the ground UEs. The channel gains from BS to UAV and UAV to UE are denoted by $\omega_{\mathbf{g}} = \omega_0 d_{bu}^{-\zeta(\varphi_{bs})}$ and $\omega_{\mathbf{h}} = \omega_0 d_{uu}^{-\zeta(\varphi_{ue})}$, respectively. Here, ω_0 is the channel power correspoing to the reference distance $d_0 = 1$ m. Therefore, the channel gains between the BS and the ith element of HRIS and that between the HRIS and i^{th} TE are characterized as

$$\mathbf{g}_{br_i}[n] = \sqrt{\omega_{\mathbf{g}}} e^{-j\frac{2\pi d_{bu}[n]}{\lambda}} \sqrt{\frac{\rho_{br}}{\rho_{br}+1}} a_R(\varphi_R[n], \eta_R[n])\, \mathbf{a}_{b,h}^H + \sqrt{\frac{1}{\rho_{br}+1}} \tilde{\mathbf{g}}_{br_i}[n], \tag{7}$$

$$\mathbf{h}_{r_{t_i}}[n] = \sqrt{\omega_h}\exp(j2\pi f_{rt} t) e^{-j\frac{2\pi d_{ut}[n]}{\lambda}} \sqrt{\frac{\rho_{rt}}{\rho_{rt}+1}} \mathbf{a}_T^{\mathrm{H}}(\varphi_T[n], \eta_T[n]) + \sqrt{\frac{1}{\rho_{rt}+1}} \tilde{\mathbf{h}}_{rt_i}[n], \tag{8}$$

where $a_R(\varphi_R, \eta_R)$ is the receive array response and given by

$$a_R(\varphi_R[n], \eta_R[n]) = \left[1, \cdots, e^{-j2\pi(L_{ax}-1)\bar{d}_x \bar{\varphi}_{AoA_{R_1}}}\right]^T \otimes \left[1, \cdots, e^{-j2\pi(L_{ay}-1)\bar{d}_y \bar{\varphi}_{AoA_{R_2}}}\right]^T, \tag{9}$$

where $L_a = L_{ax} L_{ay}$ elements, $\bar{d}_x = \frac{d_x}{\lambda}$ and $\bar{d}_y = \frac{d_y}{\lambda}$. Moreover, $\bar{\varphi}_{AoA_{R_1}} \triangleq \sin(\varphi_{AoA_{R_1}})\cos(\eta_{AoA_{R_1}}) = \frac{x_{UAV}}{\|\bar{p}\|}$ and $\bar{\varphi}_{AoA_{R_2}} \triangleq \sin(\phi_{AoA_{R_2}})\sin(\eta_{AoA_{R_2}}) = \frac{y_{UAV}}{\|\bar{p}\|}$ with $\bar{p} = [x_{UAV}, y_{UAV}, z_{UAV}]^{\mathrm{T}}$. Similarly, the transmit array response with respect to (w.r.t.) angle of departure (AoD) from the BS to the HRIS is $\mathbf{a}_{b,h}$ and $\|\mathbf{a}_{b,h}\|^2 = M$. Here, ρ_{br}, ρ_{rt} denotes the rician K-factor and each element of $\tilde{g_{br_i}}$ and $\tilde{h_{rt_i}}$ follows independent and identically distributed (i.i.d) complex Gaussian distribution with zero mean and unit variance. Similarly, $\mathbf{a}_T^{\mathrm{H}}(\varphi_T, \eta_T)$ denotes the HRIS reflect array response which is written according to (9). Also, $f_{rt} = v_d cos(\varphi_{AoA_{T_1}}) cos(\varphi_{AoA_{T_2}})/\lambda$ represents the Doppler frequency with v_d denoting the speed of the low mobility target in meters/second (m/sec), λ is the wavelength of the signal and $\varphi_{AoA_{T_1}}, \varphi_{AoAT_2}$ are azimuth and elevation angle of arrival at the RIS, respectively [54].

The transmit signal is a combination of communication and sensing symbols where $\mathbf{C}_u \in^{M\times D_u}$ is the transmit beamformer from BS to u^{th} UE, and $\mathbf{x}_u \in^{D_u\times 1}$ is the data symbol vector at u^{th} UE which satisfies $\mathbb{E}[\mathbf{x}_u \mathbf{x}_u^{\mathrm{H}}] = \mathbf{I}_{D_s}$

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and $\mathbb{E}[\mathbf{x}_u\mathbf{x}_{u'}^{\mathrm{H}}] = \mathbf{0}, u \neq u'$. $\mathbf{T}_k \in^{M\times D_k}$ is the transmit beamformer from the BS to k^{th} TE, and $\mathbf{v}_k \in^{D_k\times 1}$ is the data symbol vector at k^{th} TE which satisfies $\mathbb{E}[\mathbf{v}_k\mathbf{v}_k^{\mathrm{H}}] = \mathbf{I}_{D_s}$ and $\mathbb{E}[\mathbf{v}_k\mathbf{v}_{k'}^{\mathrm{H}}] = \mathbf{0}, k \neq k'$ and P is the transmit power. Therefore, the signal received at the uth UE is given by

$$\mathbf{y_u}[n] = \bar{\mathbf{H}}_u[n]\mathbf{C}_u\mathbf{x}_u + \sum_{i\neq u}^{U} \bar{\mathbf{H}}_u[n]\mathbf{C}_i\mathbf{x}_i + \bar{\mathbf{H}}_k[n]\sum_{k=1}^{K}\mathbf{T}_k\mathbf{v}_k + \sum_{i=1}^{L_a}\mathbf{h}_{r_{t_i}}[n]\phi_i q_a + \mathbf{n}_u, \quad (10)$$

where $\bar{\mathbf{H}}_u[n] = \mathbf{H}_{b,u}[n] + \sum_{j=1}^{L_p}\mathbf{h}_{r_{u_j}}[n]\theta_j\mathbf{g}_{br_j}[n]$ and $\bar{\mathbf{H}}_k[n] = \mathbf{H}_{b,k}[n] + \sum_{i=1}^{L_a}\mathbf{h}_{rt_i}[n]\phi_i\mathbf{g}_{br_i}[n]$. The received signal at the kth TE can be obtained as

$$\mathbf{y}_k[n] = \bar{\mathbf{H}}_k[n]\mathbf{T}_k\mathbf{v}_k + \sum_{i\neq k}^{K}\bar{\mathbf{H}}_k[n]\mathbf{T}_i\mathbf{v}_i + \bar{\mathbf{H}}_u[n]\sum_{u=1}^{U}\mathbf{C}_u\mathbf{x}_u + \sum_{i=1}^{L_a}\mathbf{h}_{r_{t_i}}[n]\phi_i q_a + \mathbf{n}_k. \quad (11)$$

The signal to interference noise ratio (SINR) at the uth UE can be expressed as

$$\gamma_u = \frac{|\bar{\mathbf{H}}_u[n]\mathbf{C}_u|^2}{\left[\sum_{i\neq u}^{U}|\bar{\mathbf{H}}_u[n]\mathbf{C}_i|^2 + \sum_{k=1}^{K}|\bar{\mathbf{H}}_k[n]\mathbf{T}_k|^2 + \sigma_a^2\sum_{i=1}^{L_a}||\mathbf{h}_{r_{t_i}}[n]\phi_i||^2 + \sigma_u^2\right]}, \quad (12)$$

Therefore, the achievable data rate at the uth UE is given by

$$R_u = \log_2(1+\gamma_u), \quad (13)$$

Similarly, SINR at the kth TE is defined as

$$\gamma_k = \frac{|\bar{\mathbf{H}}_k[n]\mathbf{T}_k|^2}{\left[\sum_{i\neq k}^{K}|\bar{\mathbf{H}}_k[n]\mathbf{T}_i|^2 + \sum_{u=1}^{U}|\bar{\mathbf{H}}_u[n]\mathbf{C}_u|^2 + \sigma_a^2\sum_{i=1}^{L_a}||\mathbf{h}_{r_{t_i}}[n]\phi_i||^2 + \sigma_k^2\right]}, \quad (14)$$

Therefore, the achievable data rate at the kth TE is given by

$$R_k = \log_2(1+\gamma_k), \quad (15)$$

A. Problem Formulation

In this subsection, we aim to maximize the achievable sum rate by jointly optimizing the phase shift matrices for the active and passive elements, and trajectory of the UAV. The optimization problem for this is formulated as

$$(\mathcal{P}_1): \max_{\mathbf{C},\mathbf{T},\theta_j,\phi_i,\iota} \sum_{u=1}^{U} R_u$$

$$\text{s.t. } 0 < \beta_i^{pas}, \beta_i^{ac} \leq 2\pi \quad (16a)$$

$$\alpha_i \leq \alpha_{max} \quad (16b)$$

$$\sum_{j=1}^{L_p}\left|h_{r_{u_j}}\theta_j g_{br_j}\right|^2 \leq \Psi_{\max} \quad (16c)$$

$$P_{a_i} \leq P_{\max_a} \quad (16d)$$

$$R_k \geq R_{min}^{thr} \quad (16e)$$

$$P \leq P_{\mathrm{T}} \quad (16f)$$

$$(1) - (2) \quad (16g)$$

where (16a) denotes that the phase shifters for passive elements are bound to have unity norm, (16b) represents the maximum amplification at the active elements of HRIS and its maximum possible transmit power is represented by (16d). (16c) enforce the passive elements of the HRIS to focus primarily on the CUs and Ψ_{max} is a predefined threshold value. Moreover, (16e) ensures the sensing performance of the targets, i.e. rate for each target should be above a threshold R_{min}^{thr} to ensure its successful detectionand (16f) restricts the transmission power. Here, $P = \sum_{u=1}^{U}||\mathbf{C}_u||^2 + \sum_{k=1}^{K}||\mathbf{T}_k||^2$ is the required transmit power at the BS. Moreover, (16g) refers to the trajectories of the UAV.

Remark 1: Note that the SR constraint (16e) can be replaced by the sensing power constraint [55]. In this case, the sensing constraint is formulated as $\bar{\gamma_k} \geq \bar{\delta}$ *where* $\bar{\gamma_k} = \frac{|\bar{\mathbf{H}}_k[n]\mathbf{T}_k|^2}{\sum_{i\neq k}^{K}|\bar{\mathbf{H}}_k[n]\mathbf{T}_i|^2+\sigma_a^2\sum_{i=1}^{L_a}||\mathbf{h}_{r_{t_i}}[n]\phi_i||^2+\sigma_k^2}$ *and* $\bar{\delta}$ *denotes the required minimum sensing power to guarantee the desired sensing performance.*

Note that, we use the *Capon beamformer* [56] for optimal solution of sensing beamformer. The sample covariance matrix of the array output is given by $\mathbf{R}_y = (\frac{1}{Q})\sum_{q=1}^{Q}\mathbf{y}_k(q)\mathbf{y}_k(q)^H$, where $q = 1, \cdots, Q$ and Q is the number of spatial samples arriving from the direction of array steering. The optimal solution that yields the capon beamformer is

$$\mathbf{t}^{\star} = \left(\mathbf{a}_R^{\mathrm{H}}(\varphi_R,\eta_R)\mathbf{R}_y^{-1}\mathbf{a}_R^{\mathrm{H}}(\varphi_R,\eta_R)\right)^{-1}\mathbf{R}_y^{-1}\mathbf{a}_R^{\mathrm{H}}(\varphi_R,\eta_R). \quad (17)$$

Accordingly, the transmit power of the active components within the H-RIS is expressed as

$$P_{a_i} = P\left(\sum_{i=1}^{L_a}|(\mathbf{g}_{br_i}[n]))\mathbf{t}\phi_i|^2\right) + \sigma^2. \quad (18)$$

As in [57], the channel training period (Ω) should be designed to occupy the minimal number of time slots such that $\Omega \geq 2U + L - 1$. Therefore, the achievable rate for t^{th} coherence interval is expressed as

$$R_u^{(n)} = \left(1 - \frac{2U+L-1}{\Omega}\right)R_u[n]. \quad (19)$$

In the case of statistical CSI, each channel is updated once per training period, with estimation performed at the start of the specified time interval. Because the optimization problem in (16) is non convex in nature, finding the globally optimal solution is quite difficult. Therefore, we employ, DTL and AO algorithm to solve problems (16), respectively. Moreover, there are few reasons to apply capon beamforming in the DDPG-based transfer learning approach rather than AO algorithm. It is because the capon beamformer entail a lot of computational power, especially in situations involving large antenna arrays or complicated signal environments. The Capon beamformer's computational load may result in longer processing times and higher resource usage when used with the AO technique, which may call for frequent updates and iterations. However, a more flexible approach to managing computing resources is provided by the DDPG-based

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transfer learning method, which would enable the effective implementation of the Capon beamformer without sacrificing performance. Additionally, accurate estimation of the covariance matrix is necessary for the Capon beamformer, but it can be difficult to acquire in dynamic and quickly changing channel conditions. The system can take advantage of the adaptability and learning capabilities of the Capon beamformer by integrating it into the DDPG-based transfer learning approach. This allows the system to continuously update and improve the beamforming strategy based on real-time feedback and observations, improving robustness to estimation errors and channel variability. The proposed solution for (16) is discussed in the subsequent section.

B. SCEE Model

In the proposed SCEE model, the channel matrices estimates for DFBS to passive elements of HRIS (BS-PHRIS), DFBS to active elements of HRIS (BS-AHRIS), DFBS to users (BS-UEs), HRIS to UEs, DFBS to TEs, and HRIS to TEs can be expressed as the sum of their estimated CSI and the associated estimation errors [58]. It is given by $\mathbf{g}_{br_j}[n] = \widehat{\mathbf{g}}_{br_j}[n] + \Delta\mathbf{g}_{br_j}[n], \mathbf{g}_{br_i}[n] = \widehat{\mathbf{g}}_{br_i}[n] + \Delta\mathbf{g}_{br_i}[n], \mathbf{H}_{b,u}[n] = \widehat{\mathbf{H}}_{b,u}[n] + \Delta\mathbf{H}_{b,u}[n], \mathbf{h}_{ru_j}[n] = \widehat{\mathbf{h}}_{ru_j}[n] + \Delta\mathbf{h}_{ru_j}[n], \mathbf{H}_{b,k}[n] = \widehat{\mathbf{H}}_{b,k}[n] + \Delta\mathbf{H}_{b,k}[n], \mathbf{h}_{rt_i}[n] = \widehat{\mathbf{h}}_{rt_i}[n] + \Delta\mathbf{h}_{rt_i}[n]$, where $\widehat{\cdot}$ denotes the estimated channels. Each SCEE vector adheres to a circularly symmetric complex Gaussian (CSCG) distribution [59] and is represented as $\mathbf{g}_{br_j}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{g}}_{br_j}[n], \mathbf{X}_{br_j} \otimes \mathbf{Y}_{br_j}\right), \mathbf{g}_{br_i}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{g}}_{br_i}[n], \mathbf{X}_{br_i} \otimes \mathbf{Y}_{br_i}\right), \mathbf{H}_{b,u}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{H}}_{b,u}[n], \mathbf{X}_{b,u} \otimes \mathbf{Y}_{b,u}\right), \mathbf{H}_{b,k}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{H}}_{b,k}[n], \mathbf{X}_{b,k} \otimes \mathbf{Y}_{b,k}\right), \mathbf{h}_{ru_j}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{h}}_{ru_j}[n], \mathbf{X}_{ru_j} \otimes \mathbf{Y}_{ru_j}\right), \mathbf{h}_{rt_i}[n] \sim \mathcal{CN}\left(\widehat{\mathbf{h}}_{rt_i}[n], \mathbf{X}_{rt_i} \otimes \mathbf{Y}_{rt_i}\right)$ where $\mathbf{X}_{br_i}, \mathbf{X}_{br_j}, \mathbf{X}_{b,u}, \mathbf{X}_{b,k}, \mathbf{X}_{rt_i}, \mathbf{X}_{ru_j}$ and $\mathbf{Y}_{br_i}, \mathbf{Y}_{br_j}, \mathbf{Y}_{b,u}, \mathbf{Y}_{b,k}, \mathbf{Y}_{rt_i}, \mathbf{Y}_{ru_j}$ are the covariance matrices at the receiver and transmitter, respectively. The approximate channel representation at BS is given by $\widehat{\mathbf{H}}_u[n] \triangleq \widehat{\mathbf{H}}_{b,u}[n] + \sum_{j=1}^{L_p} \widehat{\mathbf{h}}_{ru_j}[n]\theta_j\widehat{\mathbf{g}}_{br_j}[n]$ and $\widehat{\mathbf{H}}_k[n] \triangleq \widehat{\mathbf{H}}_{b,k}[n] + \sum_{i=1}^{L_a} \widehat{\mathbf{h}}_{rt_i}[n]\phi_i\widehat{\mathbf{g}}_{br_i}[n]$.[1] Utilizing the SCEE model, the signal received at u^{th} UE can be expressed as (20), shown at the bottom of the next page. Similar to (20), the signal received for k^{th} TE is obtained.

The achievable data rate at the uth CU can be expressed as

$$R_{u_{scee}}(\mathbf{C}, \theta_j) = \log_2|\mathbf{I} + \widehat{\mathbf{H}}_u\mathbf{C}_u\mathbf{C}_u^{\mathrm{H}}\widehat{\mathbf{H}}_u^{\mathrm{H}}\mathbf{J}_u{}^{-1}|, \tag{21}$$

where $\bar{\mathbf{C}} = \sum_{j=1}^{U}\mathbf{C}_j\mathbf{C}_j^{\mathrm{H}} \in \mathbb{C}^{M\times M}$, $\bar{\mathbf{T}} = \sum_{i=1}^{K}\mathbf{T}_i\mathbf{T}_i^{\mathrm{H}} \in \mathbb{C}^{M\times M}$, and $\mathbf{J}_u \triangleq \mathbb{E}\{\mathbf{N}_u\mathbf{N}_u^{\mathrm{H}}\}$ with N_u denoting the sum of interference and noise power at the uth CU, depicted in (22), shown at the bottom of the next page.

Lemma 1: For $\mathbf{H} \sim \mathcal{CN}(\widehat{\mathbf{H}}, \mathbf{X} \otimes \mathbf{Y})$, $\mathbb{E}[\mathbf{HAH}^{\mathrm{H}}] = \widehat{\mathbf{H}}\mathbf{A}\widehat{\mathbf{H}}^{\mathrm{H}} + \mathrm{Tr}(\mathbf{AX}^{\mathrm{T}})\mathbf{Y}$ *and* $\mathbb{E}[\mathbf{H}^{\mathrm{H}}\mathbf{AH}] = \widehat{\mathbf{H}}^{\mathrm{H}}\mathbf{A}\widehat{\mathbf{H}} + \mathrm{Tr}(\mathbf{YA})\mathbf{X}^{\mathrm{T}}$.

Proof: Refer to [60]. ■

[1]In some parts of the paper, for brevity, the time slot index $[n]$ is omitted where context allows.

Using Lemma 1, $\mathbf{J}_u$ in (22), can be rewritten as

$$\begin{aligned}\mathbf{J}_u &= \sum_{j\neq u}^{U}\Big\{\widehat{\mathbf{H}}_u\mathbf{C}_j\mathbf{C}_j^{\mathrm{H}}\widehat{\mathbf{H}}_u^{\mathrm{H}} + \widehat{\mathbf{H}}_k\bar{\mathbf{T}}\widehat{\mathbf{H}}_k^{\mathrm{H}} + \mathrm{Tr}(\bar{\mathbf{C}}\mathbf{X}_{b,u}^{\mathrm{T}})\mathbf{Y}_{b,u} + \mathrm{Tr}\{\theta_j \\ &\times[\widehat{\mathbf{g}}_{br_j}\bar{\mathbf{C}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}} + \mathrm{Tr}(\bar{\mathbf{C}}\mathbf{X}_{br_j}^{\mathrm{T}})\mathbf{Y}_{br_j}]\theta_j^{\mathrm{H}}\mathbf{X}_{ru_j}^{\mathrm{T}}\}\mathbf{Y}_{ru_j} + \mathrm{Tr}(\bar{\mathbf{C}}\mathbf{X}_{br_j}^{\mathrm{T}}) \\ &\times \widehat{\mathbf{h}}_{ru_j}\theta_j\mathbf{Y}_{br_j}\theta_j{}^{\mathrm{H}}\widehat{\mathbf{h}}_{ru_j}^{\mathrm{H}} + \mathrm{Tr}(\bar{\mathbf{T}}\mathbf{X}_{b,k}^{\mathrm{T}})\mathbf{Y}_{b,k} + \mathrm{Tr}(\bar{\phi}_i\mathbf{X}_{rt_i}^{\mathrm{T}})\mathbf{Y}_{rt_i} \\ &\times\sigma_{q_a}^2\mathbf{I}_{L_a} + \mathrm{Tr}(\bar{\mathbf{T}}\mathbf{X}_{br_i}^{\mathrm{T}})\widehat{\mathbf{h}}_{rt_i}\phi_i\mathbf{Y}_{br_i}\phi_i{}^{\mathrm{H}}\widehat{\mathbf{h}}_{rt_i}^{\mathrm{H}} + \mathrm{Tr}\{\phi_i[\widehat{\mathbf{g}}_{br_i}\bar{\mathbf{T}}\widehat{\mathbf{g}}_{br_i}^{\mathrm{H}} \\ &+ \mathrm{Tr}(\bar{\mathbf{T}}\mathbf{X}_{br_i}^{\mathrm{T}})\mathbf{Y}_{br_i}]\phi_i^{\mathrm{H}}\mathbf{X}_{rt_i}^{\mathrm{T}}\}\mathbf{Y}_{rt_i} + \widehat{\mathbf{h}}_{rt_i}\bar{\phi}_i\widehat{\mathbf{h}}_{rt_i}^{\mathrm{H}}\sigma_{q_a}^2\mathbf{I}_{L_a} + \sigma_u^2\mathbf{I}\Big\}.\end{aligned} \tag{23}$$

For simplicity, it is assumed that $\Delta\mathbf{g}_{br_j}[n]$, $\Delta\mathbf{g}_{br_i}[n]$, $\Delta\mathbf{h}_{ru_j}[n]$, $\Delta\mathbf{h}_{rt_i}[n]$, $\Delta\mathbf{H}_{b,k}[n]$ and $\Delta\mathbf{H}_{b,u}[n]$ obey the i.i.d CSCG distribution having zero-mean and variances $\sigma_{br_j}^2$, $\sigma_{br_i}^2$, $\sigma_{ru_j}^2$, $\sigma_{rt_i}^2$, $\sigma_{b,k}^2$ and $\sigma_{b,u}^2$, respectively. It is assumed that $\mathbf{X}_{br_j} = \mathbf{I}, \mathbf{Y}_{br_j} = \sigma_{br_j}^2\mathbf{I}, \mathbf{X}_{b,u} = \mathbf{I}, \mathbf{Y}_{b,u} = \sigma_{b,u}^2\mathbf{I}, \mathbf{X}_{ru_j} = \mathbf{I}$ and $\mathbf{Y}_{ru_j} = \sigma_{ru_j}^2\mathbf{I}$, $\mathbf{X}_{br_i} = \mathbf{I}, \mathbf{Y}_{br_i} = \sigma_{br_i}^2\mathbf{I}, \mathbf{X}_{b,u} = \mathbf{I}, \mathbf{Y}_{b,u} = \sigma_{b,u}^2\mathbf{I}$, $\mathbf{X}_{b,k} = \mathbf{I}, \mathbf{Y}_{b,k} = \sigma_{b,k}^2\mathbf{I}$. Eventually, (23) is rewritten as

$$\begin{aligned}\mathbf{J}_u &= \widehat{\mathbf{H}}_u\bar{\mathbf{C}}\widehat{\mathbf{H}}_u^{\mathrm{H}} + \sigma_{b,u}^2\,\mathrm{Tr}(\bar{\mathbf{C}})\mathbf{I} + \sigma_{br_j}^2\,\mathrm{Tr}(\bar{\mathbf{C}})\widehat{\mathbf{h}}_{ru_j}\widehat{\mathbf{h}}_{ru_j}^{\mathrm{H}} + \sigma_{ru_j}^2 \\ &\mathrm{Tr}(\widehat{\mathbf{g}}_{br_j}\bar{\mathbf{C}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}})\mathbf{I} + \sigma_{br_j}^2\sigma_{ru_j}^2L_p\mathrm{Tr}(\bar{\mathbf{C}})\mathbf{I} + \sigma_{q_a}^2\mathbf{I}_{L_a}\sigma_{rt_i}^2\bar{\Phi} + \sigma_u^2\mathbf{I} \\ &+ \widehat{\mathbf{h}}_{rt_i}\bar{\phi}_i\widehat{\mathbf{h}}_{rt_i}^{\mathrm{H}}\sigma_{q_a}^2\mathbf{I}_{L_a} + \widehat{\mathbf{H}}_k\bar{\mathbf{T}}\widehat{\mathbf{H}}_k^{\mathrm{H}} + \sigma_{br_i}^2\,\mathrm{Tr}(\bar{\mathbf{T}})\widehat{\mathbf{h}}_{rt_i}\widehat{\mathbf{h}}_{rt_i}^{\mathrm{H}} + \sigma_{b,k}^2 \\ &\times\mathrm{Tr}(\bar{\mathbf{T}})\mathbf{I} + \sigma_{rt_i}^2\mathrm{Tr}(\widehat{\mathbf{g}}_{br_i}\bar{\mathbf{T}}\widehat{\mathbf{g}}_{br_i}^{\mathrm{H}})\mathbf{I} + \sigma_{br_i}^2\sigma_{rt_i}^2L_p\mathrm{Tr}(\bar{\mathbf{C}})\mathbf{I}.\end{aligned} \tag{24}$$

The achievable rate for the kth TE can be written as

$$R_k(\mathbf{T}, \phi_i) = \log_2|\mathbf{I} + \widehat{\mathbf{H}}_k\mathbf{T}_v\mathbf{T}_v^{\mathrm{H}}\widehat{\mathbf{H}}_k^{\mathrm{H}}\mathbf{J}_k{}^{-1}|, \tag{25}$$

where $\mathbf{J}_k{}^{-1}$ is the sum of interference and noise power at the kth TE. The proof is omitted due to the paucity of space.

III. Designing DRL in H-RIS Assisted UAV Environment

DRL merges deep neural networks (DNNs) and RL to address Markov Decision Processes (MDP) characterized by extensive action and state spaces, which pose challenges for traditional RL methods [61], [62]. The MDP framework can be described using a tuple S, A, P, R, where S denotes the state space, which consists of observations from the network environment. A stands for the set of available actions [63]. The state transition probability, denoted as P, represents the likelihood distribution of the next state s^{t+1}, given the current action A, considered in the present state s^t. The immediate reward R assesses the quality of the state-action pair (s^t, a^t) and provides a numerical evaluation $r^t(s^t, a^t)$. The BS function as an agent in the RIS assisted UAV system. This agent engages with its surroundings to determine the best nearly optimal policy to maximize the overall sum-rate at the users. Utilizing the fundamental concept of MDP, we outline S, A and R to address the issue presented in (16) as described below.

- State Space: The state space S is described as a vector that encompasses all the details regarding the environment. In statistical CSI, the s^t includes the current HRIS

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phase shift matrix's real and imaginary parts. The latter part of the state vector is formed by the real and imaginary parts of the precoding matrix. But, when considering instantaneous CSI, the state vector s^t is expanded to include an additional element: the set of the channel coefficient's real and imaginary parts.

- Action space: Consider an action a^t within the action space A. Upon receiving state s^t from the environment, the learning agent takes action a^t at time step t. We define an action as follows: $A = \{\theta_j, \phi_i, \iota\}$.
- Reward: Once the learning agent carries out an action a^t, the environment provides an immediate feedback in the form of a reward, labeled as r^t. This reward signifies the efficacy of the chosen action. The immediate reward can be expressed as $r^t = R_u^t$. Let $R^t = \sum_{j=t}^{\infty} \upsilon^{j-t} r^t$ be the cumulative discounted reward over the long term, where $\upsilon \in (0, 1)$ symbolizes the factor by which rewards are discounted over time.

Moreover, DDPG, a DRL based approach which is built upon the actor-critic algorithm, is employed to develop an optimal policy for scenarios involving continuous states and actions. The actor network is tasked with learning the best policy for actions in a continuous space, whereas the critic network focuses on approximating the actual Q-table to mitigate the challenges posed by high-dimensional states. RL offers a solution for determining the best policy $\pi^\star : S \rightarrow A$, which assigns an action $a^t \in A$ to each state $s_t \in S$ in a way that maximizes the value function $V(s)$. When dealing with small

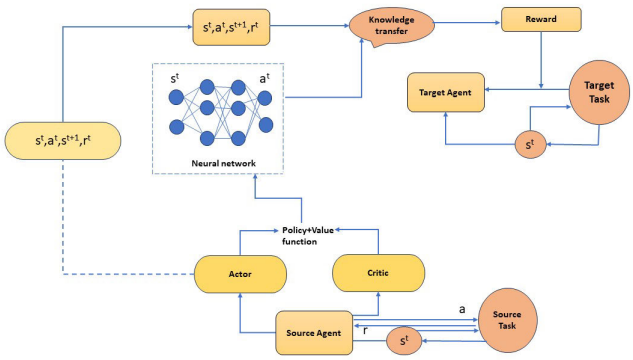


Fig. 2. Depiction of DDPG based transfer learning.

and finite state and action spaces, the Q-learning algorithm, as described in [42], allows us to find the optimal action $a^{t\star} = \arg\max_{a \in A} Q(s^t, a)$ for each state, aiming to maximize the Q-value function. The actor network receives state and action as inputs and generates actions for interacting with the environment, while the critic network assesses action performance and decides on the actions the actor network should take in the next state to achieve the highest Q-value. The loss function L is defined as

$$L = \frac{1}{B} \sum_{t=1}^{B} (y^t - Q(s^t, a^t | \varsigma^Q))^2, \tag{26}$$

where y^t is expressed as $y^t = r^t + \gamma Q'(s^{t+1}, \mu'(s^{t+1}|\varsigma^{\mu'})|\varsigma^{Q'})$. Here γ, ς_Q and ς^μ are the

$$\begin{aligned}
\mathbf{y}_u = & \left[\left(\widehat{\mathbf{H}}_{b,u} + \Delta\mathbf{H}_{b,u}\right) + \sum_{j=1}^{L_p} \left(\widehat{\mathbf{h}}_{ru_j} + \Delta\mathbf{h}_{ru_j}\right) \theta_j \left(\widehat{\mathbf{g}}_{br_j} + \Delta\mathbf{g}_{br_j}\right)\right] \sum_{u=1}^{U} \mathbf{C}_u \mathbf{x}_u + \left[\left(\widehat{\mathbf{H}}_{b,k} + \Delta\mathbf{H}_{b,k}\right)\right. \\
& \left. + \sum_{i=1}^{L_a} \left(\widehat{\mathbf{h}}_{rt_i} + \Delta\mathbf{h}_{rt_i}\right) \phi_i \left(\widehat{\mathbf{g}}_{br_i} + \Delta\mathbf{g}_{br_i}\right)\right] \sum_{k=1}^{K} \mathbf{T}_k \mathbf{v}_k + \sum_{i=1}^{L_a} \left(\widehat{\mathbf{h}}_{r_{t_i}}[n] + \Delta\mathbf{h}_{r_{t_i}}[n]\right) \phi_i q_a + \mathbf{n}_u, \\
= & \widehat{\mathbf{H}}_u \mathbf{C}_u \mathbf{x}_u + \underbrace{\widehat{\mathbf{H}}_u \sum_{i \neq u}^{U} \mathbf{C}_u \mathbf{x}_u}_{\text{IUI}} + \underbrace{\left(\Delta\mathbf{H}_{b,u} + \sum_{j=1}^{L_P} \left(\widehat{\mathbf{h}}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j} + \Delta\mathbf{h}_{ru_j} \theta_j \widehat{\mathbf{g}}_{br_j} + \Delta\mathbf{h}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j}\right)\right) \sum_{u=1}^{U} \mathbf{C}_u \mathbf{x}_u}_{\text{SCEE at u-th CU}} \\
& + \underbrace{\widehat{\mathbf{H}}_t \sum_{k=1}^{K} \mathbf{T}_k \mathbf{v}_k + \left(\Delta\mathbf{H}_{b,k} + \sum_{i=1}^{L_a} \left(\widehat{\mathbf{h}}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i} + \Delta\mathbf{h}_{rt_i} \phi_i \widehat{\mathbf{g}}_{br_i} + \Delta\mathbf{h}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i}\right)\right) \sum_{k=1}^{K} \mathbf{T}_k \mathbf{v}_k}_{\text{SCEE at k-th target}} + \underbrace{\widehat{\mathbf{h}}_{rt_i} \phi_i q_a + \Delta\mathbf{h}_{rt_i} \phi_i q_a + \mathbf{n}_u}_{\text{AEN+AWGN}},
\end{aligned} \tag{20}$$

$$\begin{aligned}
& \mathbf{J}_u \\
& = \mathbb{E}\left\{\sum_{j \neq u}^{U} \widehat{\mathbf{H}}_u \mathbf{C}_j \mathbf{C}_j^{\mathrm{H}} \widehat{\mathbf{H}}_u^{\mathrm{H}}\right\} + \mathbb{E}\left\{\Delta\mathbf{H}_{b,u} \bar{\mathbf{C}} \Delta\mathbf{H}_{b,u}^{\mathrm{H}}\right\} + \mathbb{E}\left\{(\Delta\mathbf{h}_{ru_j} \theta_j \widehat{\mathbf{g}}_{br_j})\bar{\mathbf{C}}(\Delta\mathbf{h}_{ru_j} \theta_j \widehat{\mathbf{g}}_{br_j})^{\mathrm{H}}\right\} + \mathbb{E}\left\{(\widehat{\mathbf{h}}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j})\bar{\mathbf{C}}(\widehat{\mathbf{h}}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j})^{\mathrm{H}}\right\} \\
& + \mathbb{E}\left\{(\Delta\mathbf{h}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j})\bar{\mathbf{F}}(\Delta\mathbf{h}_{ru_j} \theta_j \Delta\mathbf{g}_{br_j})^{\mathrm{H}}\right\} + \mathbb{E}\left\{\widehat{\mathbf{H}}_k \bar{\mathbf{T}} \widehat{\mathbf{H}}_k^{\mathrm{H}}\right\} + \mathbb{E}\left\{\Delta\mathbf{H}_{b,k} \bar{\mathbf{T}} \Delta\mathbf{H}_{b,k}^{\mathrm{H}}\right\} + \mathbb{E}\left\{(\Delta\mathbf{h}_{rt_i} \phi_i \widehat{\mathbf{g}}_{br_i})\bar{\mathbf{T}}(\Delta\mathbf{h}_{rt_i} \phi_i \widehat{\mathbf{g}}_{br_i})^{\mathrm{H}}\right\} \\
& + \mathbb{E}\left\{(\widehat{\mathbf{h}}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i})\bar{\mathbf{T}}(\widehat{\mathbf{h}}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i})^{\mathrm{H}}\right\} + \mathbb{E}\left\{(\Delta\mathbf{h}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i})\bar{\mathbf{T}}(\Delta\mathbf{h}_{rt_i} \phi_i \Delta\mathbf{g}_{br_i})^{\mathrm{H}}\right\} + \mathbb{E}\left\{\Delta\mathbf{h}_{rt_i} \bar{\phi}_i \Delta\mathbf{h}_{rt_i}^{\mathrm{H}}\right\} \sigma_{q_a}^2 \mathbf{I}_{L_a} \\
& + \mathbb{E}\left\{\widehat{\mathbf{h}}_{rt_i} \bar{\phi}_i \widehat{\mathbf{h}}_{rt_i}^{\mathrm{H}}\right\} \sigma_{q_a}^2 \mathbf{I}_{L_a} + \sigma_u^2 \mathbf{I},
\end{aligned} \tag{22}$$

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discount factor, networks parameters of critic and actor network, respectively. The update of the actor network's gradient is defined as

$$\nabla_{\varsigma_\mu} J \approx \frac{1}{B}\sum_{t=1}^{B} \nabla_a Q\left(s, a \middle| \varsigma^Q\right)|_{s=s^t, a=\mu(s^t)} \nabla_{\varsigma^\mu}\mu(s|\varsigma^\mu)|_{s=s^t}. \tag{27}$$

The weights of the target network are adjusted at a regular frequency τ. The weight updates are represented as follows:

$$\varsigma^{Q'} \leftarrow \tau\varsigma^Q + (1-\tau)\varsigma^{Q'}, \tag{28}$$

$$\varsigma^{\mu'} \leftarrow \tau\varsigma^\mu + (1-\tau)\varsigma^{\mu'}. \tag{29}$$

IV. Integration of DDPG With Transfer Learning

Transferring knowledge from previously learned tasks to the target problem can significantly reduce the learning cost, as the DRL agent continually interacts with its environment and learns numerous parameters, leading to enhanced learning efficiency and reduced training time. Transfer learning [64], [65] is commonly employed to address the aforementioned issues, offering the opportunity to enhance a DRL [66] agent by leveraging insights and expertise from a relevant task. Additionally, another rationale for embracing transfer learning is its capacity to diminish the volume of data required for training on the target task. As shown in Fig. 2, it considers a pre-trained model which in our case is the DDPG model and adapting to the target task. The environment in this context is the system framework in which the process takes place. It includes the computational resources, the algorithms used for training and transferring learning, etc. for evaluating model performance. Therefore, for a given source domain (X_s), a source task (T_s), a source agent (A_s), a target domain (X_t), and a target task (T_t), transfer learning is used to design target agent (A_t) with the knowledge gained from (X_s) and (T_s) where $(X_s \neq X_t)$ or $(T_s \neq T_t)$. It is worth emphasizing that when the state and action spaces of pre-trained DDPG agent differ from that of the new TL agent, the existing Q network can be adapted by adjusting the input and output dimensions. This can involve techniques like adding extra network neurons with initial parameters or randomly removing some neurons. Consequently, rather than starting the training process from the beginning, the new task agent has the potential to develop an improved policy in less time. Furthermore, because the parameters of source agent can be reused, there is no requirement for target agent to start learning these parameters anew using an extensive data set. To leverage both pre-existing knowledge and newly acquired data, hybrid value $Q^\star(s^t, a^t)$ is formulated to train target agent A_t, which is defined as follows. To solve (16), initially we convert it to a more tractable form. Subsequently, we divide it into two separate sub-problems: a) deals with the transmit beamformer design at the BS and b) deals with the reflection matrix. We then employ an AO technique to iteratively and independently resolve these sub-problems until a point of convergence.

$$Q^\star\left(s^t, a^t\right) = \upsilon^t Q\left(s^t, a^t \mid \varsigma^Q\right) + \left(1-\upsilon^t\right) Q'\left(s^t, a^t \mid \varsigma^{Q'}\right), \tag{30}$$

Algorithm 1 DRL-Based Trajectory and phase Shift Optimization

```
Initialize the trajectory and phase shift environment
Initialize source agent: Q(s^t, a^t|ς^Q), μ(s|ς_μ)
Pre-train trajectory and phase shift source agent As_S
  Initialize target trajectory and phase shift agent: Q'(s^t, a^t | ς^{Q'}), μ'(s|ς'_μ)
Initialize replay buffer υ
for e = 1, ··· , E do:
    Obtain initial UAV position and phase state
    for t = 1, ··· , T do:
        Sample channel realizations using Λ
        Take combined action a^t based on current trajectory and phase shift policy, observe next UAV position s^{t+1} and phase state
        Store transition tuples in replay buffer
        Sample a mini-batch of transitions from replay buffer
        Compute target Q*(s^t, a^t) using (30)
        Update target network Q'(s^t, a^t | ς^{Q'}) and μ'(s|ς'_μ) based on (28) and (29)
    end for
end for
```

where $\upsilon \in [0,1]$ denotes the transfer rate to which the source agent influences the target agent. The description of DTL based method is showcased in **Algorithm 1**.

V. Benchmark Scheme

A. MMSE Transformation

To address the non-convex challenge, we use MMSE approach, converting it into a more straightforward equivalent version. Notably, MMSE incorporates a linear decoding matrix $\mathbf{Q}_u, \forall u$ and estimats the signal at each UE. The estimated signal at u^{th} UE is $\hat{\mathbf{s}}_u = \mathbf{Q}_u^{\mathrm{H}}\mathbf{y}_u, \forall u$. Thererfore, the mean-square error (MSE) matrix for the uth UE is

$$\begin{aligned}\mathbf{E}_u(\mathbf{Q}_u, \mathbf{C}_u, \boldsymbol{\theta_j}) &= \mathbb{E}\{(\hat{\mathbf{s}}_u - \mathbf{s}_u)(\hat{\mathbf{s}}_u - \mathbf{s}_u)^{\mathrm{H}}\}, \\ &= \mathbf{Q}_u^{\mathrm{H}}\mathbb{E}\{\bar{\mathbf{H}}_u\bar{\mathbf{C}}\bar{\mathbf{H}}_u^{\mathrm{H}}\}\mathbf{Q}_u - \mathbf{Q}_u^{\mathrm{H}}\mathbb{E}\{\bar{\mathbf{H}}_u\}\mathbf{C}_u \\ &\quad - \mathbf{C}_u^{\mathrm{H}}\mathbb{E}\{\bar{\mathbf{H}}_u^{\mathrm{H}}\}\mathbf{Q}_u + \sigma_u^2\mathbf{Q}_u^{\mathrm{H}}\mathbf{Q}_u + \mathbf{I}. \end{aligned} \tag{31}$$

Adding a group of auxiliary variables $\mathbf{W} \triangleq \{\mathbf{W}_u \succeq 0, \forall u \in \mathcal{U}\}$ and $\mathbf{Q} \triangleq \{\mathbf{Q}_u, \forall u \in \mathcal{U}\}$, (21) is further expressed as

$$\begin{aligned}&\hat{R}_{u_{scee}}(\mathbf{C}, \theta_j, \mathbf{Q}, \mathbf{W}) \\ &= \sum_{u=1}^{U}\Big\{\mathrm{Tr}(\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\mathbf{V}_u\mathbf{Q}_u) \\ &\quad - \mathrm{Tr}(\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\widehat{\mathbf{H}}_u\mathbf{C}_u) + \mathrm{Tr}(\mathbf{W}_u) - \mathrm{Tr}(\mathbf{W}_u\mathbf{C}_u^{\mathrm{H}}\widehat{\mathbf{H}}_u^{\mathrm{H}}\mathbf{Q}_u) \\ &\quad + \mathrm{Tr}(\sigma_u^2\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\mathbf{Q}_u) - \log_2|\mathbf{W}_u|\Big\}, \end{aligned} \tag{32}$$

where $\mathbf{V}_u \triangleq \mathbb{E}\{\widehat{\mathbf{H}}_u\bar{\mathbf{C}}\widehat{\mathbf{H}}_u^{\mathrm{H}}\}$ which is procured as similar to $\mathbf{J}_u$. Due to space constraints, the derivation is omitted. Now, the optimization problem $\mathcal{P}_1$ in (16) is re-framed as

$$\begin{aligned}(\mathcal{P}_2): \max_{\mathbf{C},\mathbf{T},\theta_j,\phi_j,\mathbf{Q},\mathbf{W},\iota} &\sum_{u=1}^{U}\hat{R}_u(\mathbf{C}, \theta_j, \mathbf{Q}, \mathbf{W}) \\ \text{s.t. } &(16a)-(16g), \end{aligned} \tag{33a}$$

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Since, the optimization problem $\mathcal{P}_2$ in (33) involves a large number of optimization variables than $\mathcal{P}_1$. We use an AO framework which separates the transformed problem in (33) into three sub-problems: precoder design at BS, HRIS phase shift matrix design and UAV trajectory design (Section V-D).

B. Transmit Beamforming Design

Initially, we solve $\mathbf{Q}$ and $\mathbf{W}$ with specified values for $\mathbf{C}$ and θ_j. The optimal $\mathbf{Q}_u$ is obtained by setting the first-order derivative of (31) w.r.t $\mathbf{Q}_u$ to zero,

$$\mathbf{Q}_u^* = \arg\min_{\mathbf{Q}_u} \mathbf{E}_u = (\mathbf{V}_u + \sigma_u^2\mathbf{I})^{-1}\widehat{\mathbf{H}}_u\mathbf{C}_u. \tag{34}$$

Next, we substitute $\mathbf{Q}_u^*$ in (31) to calculate $\mathbf{E}_u$ and $\mathbf{W}_u$ as

$$\mathbf{E}_u^* = \mathbf{I} - \mathbf{C}_u^{\mathrm{H}}\widehat{\mathbf{H}}_u^{\mathrm{H}}\mathbf{Q}_u\,, \qquad \mathbf{W}_u^* = (\mathbf{E}_u^*)^{-1}. \tag{35}$$

Similarly, by addressing the rate constraint at the TE with the MMSE method, the problem in (36a) turns into convex w.r.t $\mathbf{C}, \mathbf{T}$, as

$$(\mathcal{P}_{2_{a_1}}): \min_{\mathbf{C},\mathbf{T},\mathbf{Q},\mathbf{W}} \sum\nolimits_{u=1}^{U} \hat{R}_u(\mathbf{C},\mathbf{Q},\mathbf{W}|\theta_j) \tag{36a}$$

$$\text{s.t. (16d)} - \text{(16g)}, \tag{36b}$$

The problem presented in (36a) is a convex since its objective function displays convexity with its affine constraints in $\boldsymbol{C}$ and $\boldsymbol{T}$. Its solution can be achieved using either Lagrangian dual approach or standard CVX toolbox. Provided with an initial feasible power distribution $\boldsymbol{C}(0)\forall u$ and $\boldsymbol{T}(0)\forall k$ the problem (36a) undergoes multiple iterations until the objective function stabilizes. Moreover, the non-convex rate constraint for k^{th} target follows the similar approach.

C. HRIS Beamforming Design

In this subsection, we solve the hybrid phase shift design for the obtained $\mathbf{C}, \mathbf{T}, \mathbf{Q}_\mathbf{u}^\star, \mathbf{W}^\star$ as

$$(\mathcal{P}_{2_{b_1}}): \min_{\theta_j} \; h(\theta_j)$$

$$\text{s.t. (16a)} - \text{(16e)}, \tag{37a}$$

where $h(\theta_j) = (e^{j\beta_j^{pas}})^{\mathrm{H}}\mathbf{A}e^{j\beta_j^{pas}} + 2\mathrm{Re}\{(e^{j\beta_j^{pas}})^{\mathrm{H}}\mathbf{f}\}$, $\mathbf{A} = \mathbf{A}_0+\mathbf{A}_1+\mathbf{A}_2+\mathbf{A}_3$, $\mathbf{f} = \big[[\mathbf{F}]_{1,1},\ldots,[\mathbf{F}]_{L,L}\big]$, $\mathbf{A}_0 = \mathbf{S}_0\circ\mathbf{T}_0^{\mathrm{H}}$, $\mathbf{A}_1 = \mathbf{S}_1\circ\mathbf{I}$, $\mathbf{A}_2 = \mathbf{I}\circ\mathbf{T}_1^{\mathrm{H}}$ and $\mathbf{A}_3 = \mathbf{I}\circ\mathbf{S}_2$

$$\mathbf{S}_0 = \sum\nolimits_{u=1}^{U} \widehat{\mathbf{h}}_{ru_j}\mathbf{Q}_u\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\widehat{\mathbf{h}}_{ru_j}^{\mathrm{H}}, \tag{38}$$

$$\mathbf{S}_1 = \sum\nolimits_{u=1}^{U} \sigma_{br_j}^2\,\mathrm{tr}(\bar{\mathbf{C}})\widehat{\mathbf{h}}_{ru_j}\mathbf{Q}_u\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\widehat{\mathbf{h}}_{ru_j}^{\mathrm{H}}, \tag{39}$$

$$\mathbf{S}_2 = \sigma_{br_j}^2\sigma_{ru_j}^2\,\mathrm{tr}\left(\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\mathbf{Q}_u\right)\mathrm{tr}(\bar{\mathbf{C}}), \quad \mathbf{T}_0 = \widehat{\mathbf{g}}_{br_j}\bar{\mathbf{C}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}}, \tag{40}$$

$$\mathbf{T}_1 = \sum\nolimits_{u=1}^{U} \sigma_{ru_j}^2\,\mathrm{tr}\left(\mathbf{W}_u\mathbf{Q}_u^{\mathrm{H}}\mathbf{Q}_u\right)\widehat{\mathbf{g}}_{br_j}\bar{\mathbf{C}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}}, \tag{41}$$

$$\mathbf{F} = \sum_{u=1}^{U}\left[\widehat{\mathbf{h}}_{ru_j}\mathbf{Q}_u\mathbf{W}_u\left(\mathbf{Q}_u^{\mathrm{H}}\widehat{\mathbf{H}}_{b,u}^{\mathrm{H}}\bar{\mathbf{C}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}} - \mathbf{C}_u^{\mathrm{H}}\widehat{\mathbf{g}}_{br_j}^{\mathrm{H}}\right)\right]. \tag{42}$$

We employ the SCA technique to address the surrogate function of $\mathcal{P}_{2_{b_1}}$. Using Taylor expansion, the approximation of $h(\theta_j)$ can be described as

$$\widehat{h}(\theta_j|\theta_j^r) = h(\theta_j^r) + \nabla h(\theta_j^r)^T(\theta_j - \theta_j^r) + \frac{\tau_r}{2}||\theta_j - \theta_j^r||^2, \tag{43}$$

where τ is selected to satisfy $\widehat{h}(\theta_j|\theta_j^r) \le h(\theta_j)$ and the gradient is expressed as

$$\nabla h(\theta_j^r) = 2\mathrm{Re}\{-je^{-j\beta_j^{pas^r}} \circ (\mathbf{F}e^{j\theta_j^r} + f)\}. \tag{44}$$

Thus, θ_j is updated as $\theta_j^{r+1} = \theta_j^r - \nabla h(\theta_j^r)/\tau_r$. Similarly, ϕ_i is updated.

D. UAV Trajectory Optimization

For a given phase shifts and transmit beamformer, the UAV trajectory problem is expressed as

$$(\mathcal{P}_{2_{c_1}}): \max_{\iota} \sum\nolimits_{u=1}^{U} R_u \tag{45}$$

$$s.t.\ (1), (2). \tag{46}$$

Clearly, problem (45) is not convex concerning the trajectory variables ι. Further, we determine the value of B_n [67] as

$$B_n = \frac{\wp_0 P}{\sum\limits_{i\neq u}\frac{\wp_0 P}{\|\varpi[n]-\varpi_{UEi}\|^2+z_{UAV}^2}+1}. \tag{47}$$

By defining the slack variable $A = \{A[n] = \|\varpi_{UAV}[n] - \varpi_{UE}\|^2 + z_{UAV}^2\}$, the sub-problem is written as

$$(\mathcal{P}_{2_{c_2}}): \max_{\iota} \sum\nolimits_{u=1}^{U} \log_2(1+\frac{B_n}{A[n]}) \tag{48}$$

$$s.t.\ \|\varpi_{UAV}[n] - \varpi_{UE}\|^2 + z_{UAV}^2 - A[n] \le 0,$$
$$(1), (2). \tag{49}$$

It is important to note that the optimal solution remains unchanged between problem (45) and (49) due to the new constraint. For the non-convexity of problem (49), SCA method is proposed. Let us consider ι_t as the trajectory for t^{th} iteration, and the SCA method is employed to substitute the term $\log_2\left(1+\frac{B_n}{A[n]}\right)$ with its convex lower approximation. Consequently, using the given trajectory ι_t, we can derive the first-order Taylor expansion of the item through the subsequent approximation as

$$\log_2\Big(1+\frac{B_n}{A[n]}\Big) \ge \log_2\Big(1+\frac{B_n}{A_t[n]}\Big) - \frac{B_n(A[n]-A_t[n])}{\ln 2[A_t[n]^2 + B_nA_t[n]]}, \tag{50}$$

$$\|\varpi_{UAV}[n]-\varpi_{UE}\|^2 \ge \|\varpi_{UE}\|^2 - \|\varpi_{UAV\,t}[n]\|^2 + 2[\iota_t[n]-\varpi_{UE}]^T. \tag{51}$$

Using (50) and (51), we can approximate sub problem as

$$(\mathcal{P}_{2_{c_3}}): \max_{\iota,B} \sum\nolimits_{u=1}^{U} [-\frac{B_n}{\ln 2[A_t[n]^2 + B_nA_t[n]]}A[n]] \tag{52}$$

$$s.t.\ \|\iota[n] - \varpi_{UE}\|^2 + z_{UAV}^2 - A[n] \le 0, \tag{53}$$

$$(1),\ (2). \tag{54}$$

Using this transformation, the optimization problem (52) becomes convex. This can be addressed using CVX. The specifics of the above mentioned problem is described in **Algorithm 2**. Therefore, the algorithm along with proposed solutions is outlined in **Algorithm 3**.

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Algorithm 2 Algorithm for Trajectory Design

1: Output ι_{opt}.
2: **Initialize:** Trajectory ι_0;
3: **repeat**
4: With given θ_j, ϕ_i and the slack variable $A_t[n]$, update the moving trajectory by solving problem (52);
5: Update $t = t + 1$;
6: **until** The fractional increase of the objective value is below the threshold δ;
7: **return:** ι_{opt}.

Algorithm 3 The AO Algorithm

1: Initialize $r = 0$, $\mathbf{C}^{(0)}, \mathbf{T}^{(0)}, \theta_j^{(0)}, \phi_i{}^{(0)}, \iota^{(0)}, \varepsilon$
2: Evaluate $\mathbf{Q}^{(0)}$ and $\mathbf{W}^{(0)}$ using (34) and (35), respectively
3: **repeat**
4: Obtain $\mathbf{C}^{(r+1)}$ and $\mathbf{T}^{(r+1)}$ by solving (36a) given $\theta_j^{(r)}, \phi_i^{(r)}, \mathbf{Q}^{(r)}, \mathbf{W}^{(r)}, \iota^r$
5: Obtain $\theta_j{}^{(r+1)}$ and $\phi_i{}^{(r+1)}$ by solving (37a) given $\mathbf{C}^{(r+1)}, \mathbf{T}^{(r+1)}, \mathbf{Q}^{(r)}, \mathbf{W}^{(i)}, \iota^r$
6: Obtain ι using Algorithm 2 for given $\mathbf{C}^{(r+1)}, \mathbf{T}^{(r+1)}$, θ_j^{r+1}, ϕ_i^{r+1}
7: Update $\mathbf{Q}^{(r+1)}, \mathbf{W}^{(r+1)}$ using $\mathbf{C}^{(r+1)}, \mathbf{T}^{(r+1)}, \theta_j{}^{(r+1)}, \phi_i{}^{(r+1)}$
8: $r \leftarrow r + 1$
9: **until** convergence is achieved.

E. Complexity Analysis

In this subsection, the proposed Algorithm 1 efficiency is showcased through a comparative analysis with the traditional method. The computational complexity of DTL is a function of the neural network's parameters, the batch size from the experience replay, the dimensions of the state and action spaces, the extent of fine-tuning in transfer learning, and the number of iterations during training network. Therefore, the complexity of the DTL framework is given by $\mathcal{O}(ETW(\tau \sum_{l_{m=1}}^{\mathcal{L}} l_m \times l_{m+1}))$ where $\mathcal{L}$ and l_m denotes the number of neural network layers and neurons, respectively. The total complexity of the AO algorithm is given by $\mathcal{O}(I_B UM^3 + 3I_A UL^2 N + I_C UL^2)$ where I_B, I_A, I_C denotes the number of iterations. Here, I_B represents the iterations over an optimization problem that involves beamforming vector optimization, where the computational complexity increases significantly with M, I_A denotes the iterations in determining complexity of optimizing phase shift and I_C represents the number of iterations of SCA in solving UAV trajectory. For further clarification, we also include a comparison of the complexity between the proposed transfer learning based approach and the AO-based approach is shown in Fig. 3.

VI. Simulation Results

In this section, we present simulation outcomes to corroborate the efficacy of the proposed strategies. Except where specified, we consider $La = 20$, $\Delta T = 0.1$ sec and a maximum amplification level of 10 dB. The BS is located at $(0, 0, 0)$ m, UEs are selected randomly from a 100×100 m^2 and targets are generated from the detection zone mentioned in section (II). The errors in channel estimation

TABLE III
Simulation Parameters [35], [40]

Parameters	Description	Value
M	Number of ISAC transmitter antennas	4
N	Number of UE and TE antennas	2
U, K	Number of UEs, TEs	3
L	Number of HRIS elements	40
$\sigma_u{}^2$	Noise power	-94 dBm
σ_k^2	Active element noise power	-70 dBm
P_T	Transmit power at ISAC transmitter	10 dB

TABLE IV
Configuration Parameters for DRL

Parameters	Description	Value
E	Number of episodes	1000
B	Batch-size	32
ς^μ	Learning rate of actor network	0.001
ς^Q	Learning rate of critic network	0.001
T	Number of steps	300
γ	Discount factor	0.9
W	Replay buffer	10000
τ	Update rate	{0,1}

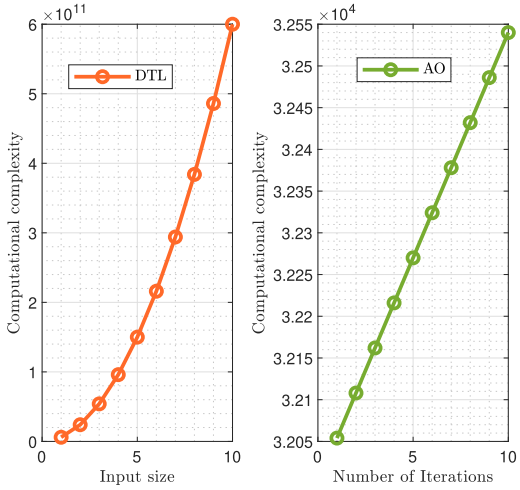


Fig. 3. Comparison of complexity analysis.

are governed by an independent and i.i.d. CSCG distribution with a zero mean. These errors have a consistent normalized mean squared error (NMSE) which is represented as $\varrho = \mathbb{E}\left[|\mathbf{g}_{br_j}[n] - \hat{\mathbf{g}}_{br_j}[n]|^2\right] / \mathbb{E}\left[|\hat{\mathbf{g}}_{br_j}|^2\right]$. The benchmark schemes for the proposed framework are as follows:

- HRIS with perfect CSI (pCSI): It means that the system has complete and accurate information about the CSI. We denote the HRIS with pCSI for AO as AO-pCSI and HRIS with pCSI for DTL as DTL-pCSI.
- Only passive RIS: For this scheme, the maximization problem is addressed by fully passive RIS ($L_a = 0$) which represents a specific instance of the HRIS.

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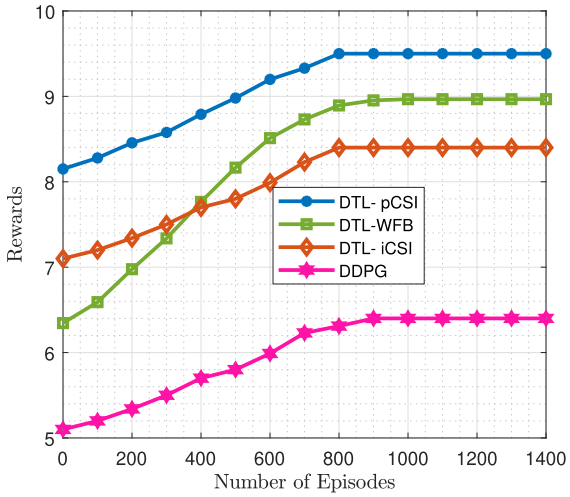


Fig. 4. Rewards versus the number of episodes.

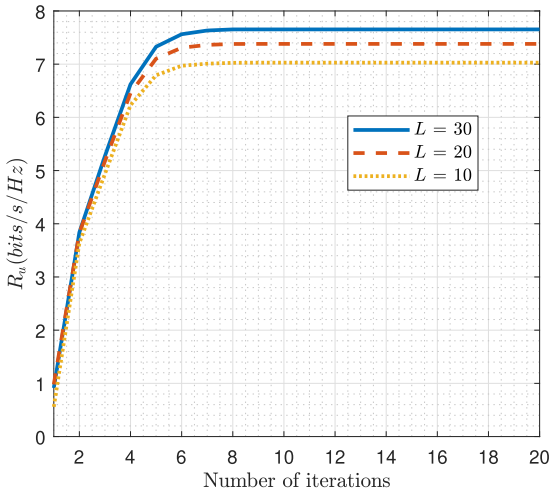


Fig. 5. R_u versus the number of iterations.

- HRIS with random phase shifts: In this scheme, the phase of HRIS elements are chosen randomly and are not subject to any optimization.
- No-RIS: This scheme considered only transmissions via direct link.
- DDPG: It is used as a benchmark algorithm which is a model-free, actor-critic algorithm leveraging the deterministic policy gradient to function in continuous action spaces.

In Fig. 4, we illustrate the rewards gained during the training process by the proposed DTL-based, DTL without fixed beamformer (DTL-WFB) and DDPG-based algorithms by varying the number of episodes. It is observed that the initial training curves for average rewards consistently increasing and then attaining a stable performance. However, as the training progresses and the networks begin to converge, there is a noticeable increase in average rewards for DTL-pCSI as compared to DTL-iCSI and DDPG. This stabilization signifies that the system has reached its optimal performance level. It is due to the fact that with pCSI, the DTL model can quickly and efficiently adapt to new environments or conditions, as it has reliable data to base its adjustments on. Specifically, the maximum rewards for DTL-pCSI, DTL-WFB, DTL-iCSI and DDPG are approximately $\{9.5, 8.9, 8.3$ and $6.4\}$ bits/s/Hz, respectively. Additionally, due to the utilization of capon beamformer, the performance of rewards is better as compared to DTL-WFB in the direction of interest while suppressing interference from other directions. When environmental parameters and interference do not vary significantly, the fixed Capon beamformer sometimes outperform adaptive or jointly optimized systems.The theoretical convergence analysis of the proposed algorithm can be analyzed according to [68], [69].

The illustration of the convergence of **Algorithm 3** with various number of iterations is shown in Fig. 5. It is observed that the performance of the proposed method with SCEE stabilizes only after 6 iterations. This rapid convergence shows the efficiency of our proposed solutions. It can be seen that for $L = 30$, the performance gain in terms of R_u improves with the increasing values of L. Specifically, when $L = 30$, the system attains a peak reward of 7.7 bits/s/Hz. Conversely, for $L = 20$ and $L = 10$, the highest rewards achieved are around 7.3 and 6.9 bits/s/Hz, respectively. Moreover, the increase in the performance of R_u is observed with an increase in L. This is due to the increase in reflecting gain at the H-RIS.

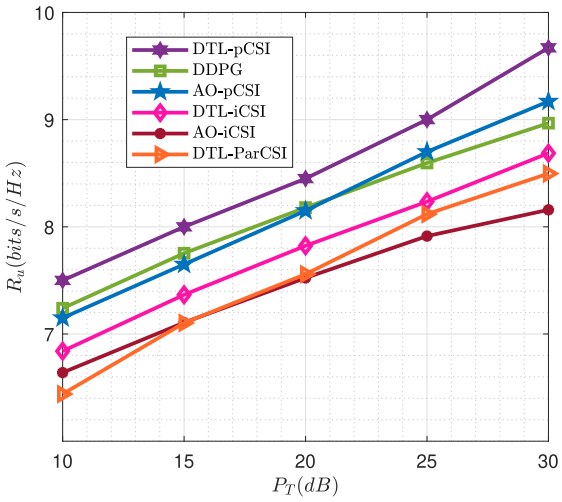


Fig. 6. R_u versus P_T (in dB).

Fig. 6 illustrates the relationship between R_u and the BS's transmit power P_T with $L = 20$ for four unique frameworks: 1) DTL-pCSI, 2) AO-pCSI, 3) DTL-iCSI and 4) AO-iCSI 5) DTL-ParCSI (partial CSI). This figure shows a progressive increment in R_u for the proposed method and the benchmark schemes with increase in P_T. The R_u of the proposed DTL approach with perfect CSI is approximately $5.33, 10.29$ and 15.38% more efficient as compared to the scenarios with AO-pCSI, DTL-iCSI, DTL-ParCSI and AO-iCSI. It is also observed that the optimized configuration of HRIS for DTL in ISAC has enhanced the communication performance by effectively managing both received and reflected signals. In scenarios where partial CSI is used, the R_u is lower due to insufficient channel information, compared to systems utilizing imperfect CSI which can adapt more effectively to channel variations through broader but less precise estimates.

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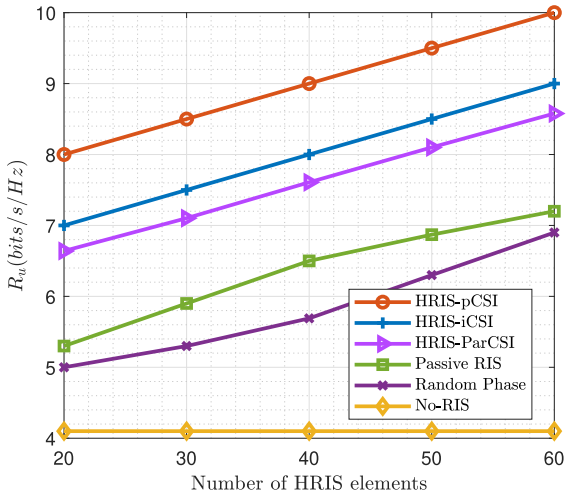


Fig. 7. R_u versus the number of HRIS elements.

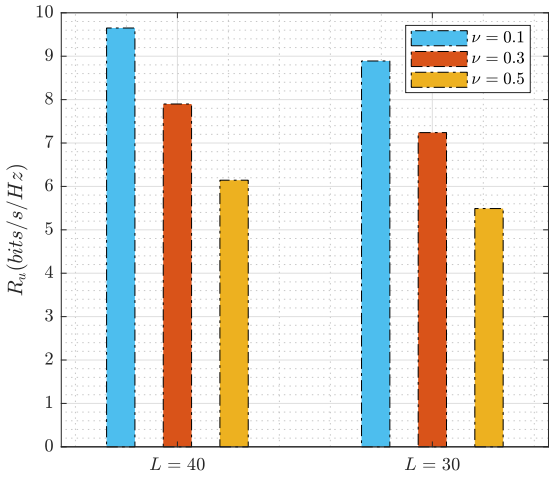


Fig. 9. R_u for various active element budget percentage.

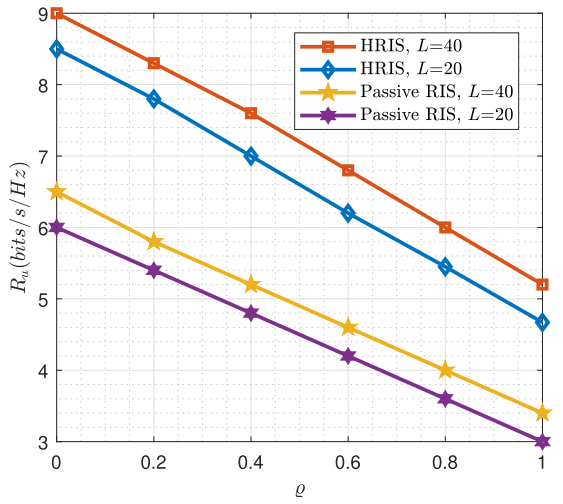


Fig. 8. R_u versus SCEE (ϱ).

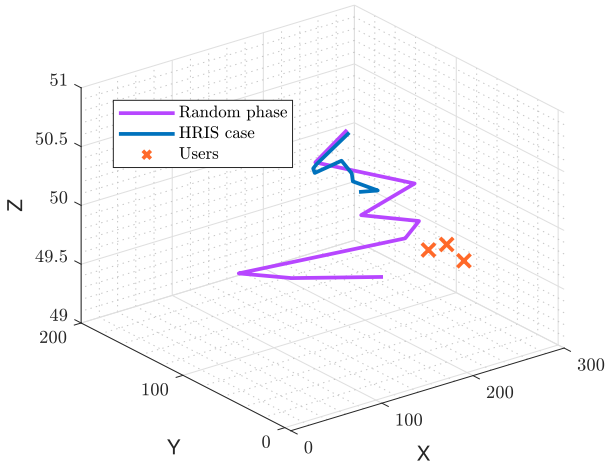


Fig. 10. The HRIS aided UAV trajectory for the proposed framework.

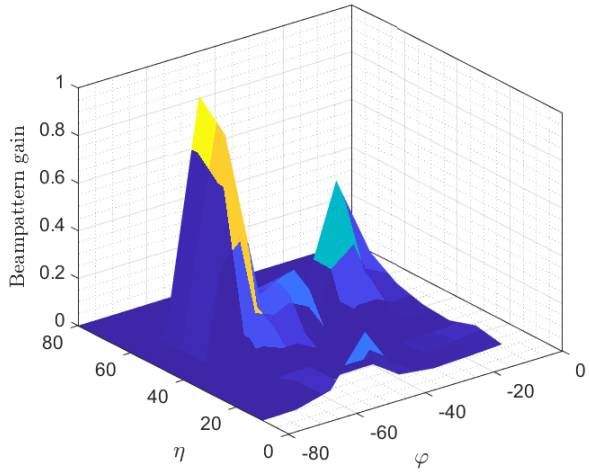


Fig. 11. Beampattern gain under iCSI case.

Fig. 7 shows the variation in R_u w.r.t. the number of HRIS elements. We compared the data rates achieved by our scheme with those of established benchmarks: pCSI, iCSI, ParCSI, passive RIS, No RIS and HRIS with random phase shifts. It can be clearly observed that the proposed framework significantly outperform the benchmark schemes with passive RIS, random phase and No-RIS case with increasing L. Furthermore, it is noteworthy that after considering SCEEs, having a higher L might not necessarily be beneficial unless they are adequately optimized. This is because the interference factor tends to rise as the count of L increases. The R_u of the proposed HRIS-pCSI is approximately $\{1.1, 2.7, 3, 3.8\}$ bits/s/Hz more efficient as compared to the scenarios with HRIS-iCSI, HRIS-ParCSI, passive RIS, random phase and no-RIS cases.

Fig. 8 describes the performance in terms of R_u against SCEE (ϱ) for different values of L. 8. It is observed that a substantial enhancement in R_u for the proposed algorithm obtained as compared to the case of passive RIS, even in the presence of iCSI. This observation highlights the efficacy of our proposed solutions. Further, it is important to emphasize that when SCEE are significant, for instance, when ϱ exceeds 0.8, it becomes evident that the performance degrades. In Fig. 9, we examine allocation of active elements at the HRIS on R_u for different L. Here ν is considered as the proportion of the budget allocated to active elements where $0 < \nu < 1$.

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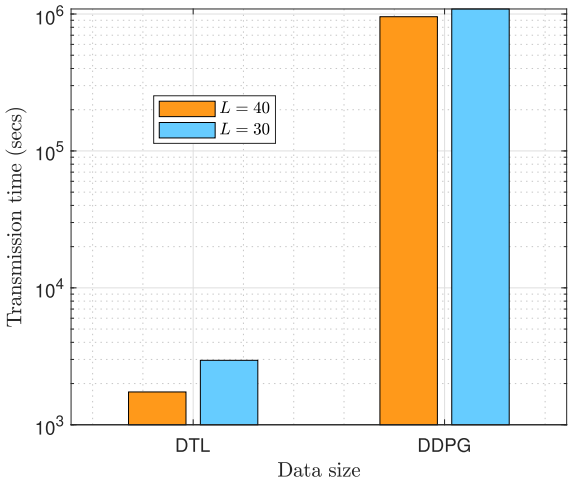


Fig. 12. Transmission time for DTL and DDPG.

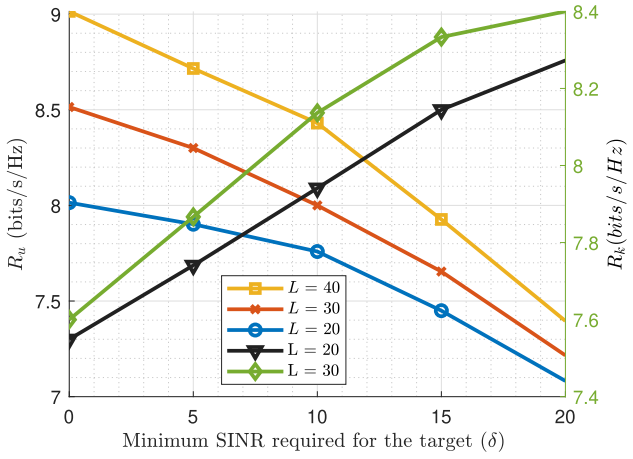


Fig. 13. Communication and target rate versus δ.

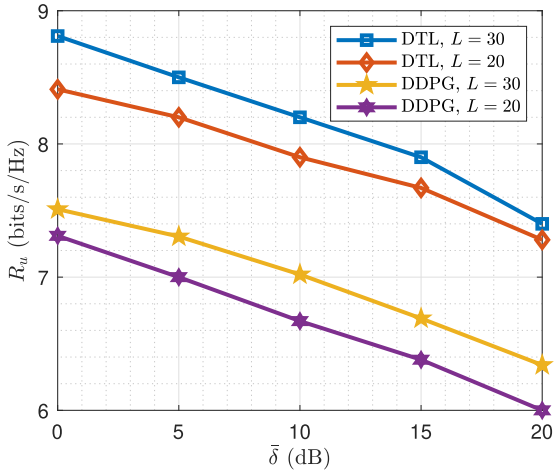


Fig. 14. Communication rate versus $\bar{\delta}$ for different number of RIS elements.

It is observed that the increasing ν, decreases R_u. This clearly shows the significant impact of allocation of active elements on the rate performance. This outcome is anticipated as a smaller ν leads to a limited number of such elements, thereby not fully utilizing the potential power amplification gain that active elements can provide.

In Fig. 10, the graph for HRIS aided UAV trajectory for the proposed framework is illustrated. It is observed that the proposed method for HRIS case appears to be the most direct and shortest path in order to reflect signals to the ground UEs. It is the optimal path which allows it to take the most energy-efficient route that begins its trajectory at the starting coordinates in order to enhance the performance. On the otherhand, for random phase, the movement of the UAV aided RIS is not predetermined but rather selected randomly. As the UAV progresses along its trajectory, despite the random phase, it systematically adjusts its course to end its journey close to the UEs location. The final segment of the UAV's route shows a deliberate approach toward the UE, suggesting a shift from a random to a targeted flight strategy.

Fig. 11 illustrates the relationship between beampattern gain across various azimuth and elevation angles. In real-world scenarios, obtaining precise target data is often challenging. Given this imprecision in target information, we introduce a resilient target sensing beampattern model to evaluate the sensing accuracy. Consequently, for k^{th} target, the sensing beampattern gain corresponding to φ and η is described in this figure. It is clear that the performance consistently diminishes as the angle error increases. Owing to the lack of precise target information in the imperfect scenario, it's beneficial to spread the sensing signal power throughout the estimated target area.

The Fig. 12 represents the transmission time for the proposed DTL and DDPG considering different RIS elements. This figure illustrates how DTL can enhance performance in communication systems by reducing transmission times. DTL is considered to reduce transmission time as compared to traditional DDPG method. It represents a significant efficiency gain, as the pre-trained model adapts and optimizes faster in the new environment. By utilizing a pre-trained model, the system needs less time to learn optimal policies, thus achieving lower transmission times more quickly compared to the case of only DDPG. Thereby, it demonstrates how knowledge from a previously trained model can effectively reduce the time needed to achieve optimal transmission strategies in a new domain.

In order to provide a more comprehensive view of the interaction between sensing and communication systems, Fig. 13 shows the trade-off between the achievable rate at the user and sensing rate at the target. This new plot provides a more direct illustration of how increasing δ affects communication performance where δ defines the minimum SINR needed to guarantee the sensing performance of the radar system. Fig. 13 illustrates the variation in R_u w.r.t. δ for different number of HRIS elements. It illustrates that as δ increases, the R_u decreases and R_k increases indicating a clear trade-off between communication and sensing performance. The decreasing nature of R_u is due to the fact that high radar SNR leads to stronger radar signals which cause more interference to communication users.

Fig. 14 shows the variation of the CR R_u with respect to $\bar{\delta}$ (i.e., the minimum sensing power required to guarantee the sensing performance of the radar system). It shows that the performance in terms of R_u decreases as $\bar{\delta}$ increases for

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both the proposed DTL and DDPG approaches. The proposed DTL consistently outperforms DDPG for all values of L demonstrating the effectiveness of DTL in achieving higher rates for the proposed framework. This also highlights the effectiveness of DTL in optimizing the trade-off between sensing and communication, making it more robust and efficient solution for ISAC systems.

VII. Conclusion

In this paper, we explored the combined transmit and phase shift beamforming along with UAV trajectory in a HRIS empowered UAV assisted joint sensing and communication scenario that accounts the channel estimation inaccuracies. We structured an optimization problem with the goal of maximizing the achievable rate. Further, we introduced a transfer learning based approach combining with DDPG to enhance the overall data transmission rate. Meanwhile, determining their nearly-optimal solution is a combinatorial challenge. To address this non-convex optimization challenge, we introduced AO algorithm of low complexity as a baseline method. Further, we considered case like DDPG, HRIS with random phase shifts and No-RIS as the benchmark schemes. Through simulation results, we showcased the efficacy of our proposed methods, with the notable dominance over baseline schemes. For example, the R_u of the proposed HRIS with perfect channel state information is approximately $1.1, 2.7, 3$ and 3.8 bits/s/Hz higher as compared to the scenarios of HRIS with imperfect CSI, passive RIS, random phase and no-RIS.

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